

Evaluating the Impact of Policy Changes on Economic Outcomes

A Counterfactual Analysis of the Federal Reserve's 2022 Monetary Tightening Cycle

Project 2: Counterfactual Analysis Project

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GitHub: github.com/albertokabore/Project-2---Evaluating-the-Impact-of-Policy-Changes-on-Economic-Outcomes

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1. Introduction

1.1 The Policy Change and Research Question

In March 2022 the Federal Open Market Committee started raising the federal funds rate from near zero, about 0.00 to 0.25 percent, up to a peak of 5.25 to 5.50 percent by mid 2023. It was the fastest pace of rate hikes in roughly four decades. The Fed's stated goal was simple: bring inflation, which had hit a 40 year high, back down toward its 2 percent target.

Research question: what effect did this rate hiking cycle have on U.S. inflation and output, compared with the path those indicators would have followed if the Fed had left rates alone?

A quick note on scope. The policy change studied here already happened, the Fed did raise rates. The hypothetical part of this project is the counterfactual alternative: what inflation and output would look like if the Fed had not raised rates. That hypothetical scenario is exactly what Sections 3 and 4 simulate and compare against what actually happened.

1.2 Motivation

Monetary policy applies to the whole economy at once, so there is no control group sitting around somewhere as “the same U.S. economy that didn't get a rate hike.” A simple before and after comparison would mix the policy's effect together with everything else happening in the economy over the same stretch of time: supply chains normalizing, energy prices swinging, pandemic era fiscal stimulus fading out. Figuring out what the policy actually did means building a counterfactual instead, a model based estimate of what inflation and output would have done without the hikes, with those other global forces stripped out as much as the data allows.

2. Data Description

2.1 Source and Nature of the Dataset

Every series in this project comes from FRED, the Federal Reserve Economic Data service run by the Federal Reserve Bank of St. Louis, a public source for U.S. and international macroeconomic data. Data were pulled using the pandas-datareader library, which reads each series' public CSV file directly. No API key is needed, and every raw pull is cached locally, so rerunning the analysis later produces the same numbers.

The full dataset runs from 2006-01-01 to 2026-08-01 at monthly frequency, 230 months of data after cleaning. It is split by date rather than randomly: a pre-intervention window (2006-01-01 to 2022-02-01, 194 months) that the model learns from, and a post-intervention window (2022-03-01 to 2025-02-01, 36 months) that is held out to compare the observed outcome against the model's counterfactual guess.

2.2 Key Variables and Economic Indicators

Variable	FRED Code	Frequency	Role
Federal funds rate	FEDFUNDS	Monthly	Policy variable (the treatment)
CPI, all items	CPIAUCSL	Monthly	→ YoY inflation (primary outcome)
PCE price index	PCEPI	Monthly	→ YoY inflation (robustness outcome)
Industrial production	INDPRO	Monthly	→ YoY growth (output outcome)
Unemployment rate	UNRATE	Monthly	Secondary outcome

Real GDP	GDPC1	Quarterly	→ YoY growth, secondary outcome
Euro area HICP	CP0000EZ19M086NEST	Monthly	Control covariate
UK CPI	GBRCPIALLMINMEI	Monthly	Control covariate
Global commodity price index	PALLFNFINDEXM	Monthly	Control covariate

The three control series come from economies and markets that were not directly subject to the Fed's decision. They let the model separate out inflation and output movements caused by global shocks, most obviously the 2021 to 2022 supply chain and energy shock, from movements caused by the Fed's own choice. That distinction matters, because otherwise the estimated effect would just pick up the shock the Fed was reacting to, not the policy itself.

2.3 Preprocessing Steps

- Index level series, CPI, PCE, industrial production, GDP, both foreign CPI controls, and commodity prices, are converted to year over year percent change. That puts every outcome on a comparable, roughly stationary scale instead of a raw, trending index level.
- All series are aligned to a common month start index. The quarterly GDP series is forward filled, up to two months, so it lines up with the monthly index.
- Short, isolated reporting gaps are forward filled, again up to two months. Rows still missing the treatment variable or the primary outcome after that get dropped. In practice this only removes the first twelve months of history, which the year over year transform consumes anyway.
- Normalizing the control covariates. Right before the model is fit, the three control covariates are z-scored, mean centered and scaled to unit variance, using statistics from the pre-intervention period only, and that same transform is then applied to the post-intervention covariates. This matters because the controls sit on very different scales (global commodity price swings are much bigger than UK CPI swings), and normalizing keeps the regression numerically stable without letting any post-intervention information leak backward into the transform.
- The train and test split is done by date, not at random. The pre-intervention period is the training set, and the post-intervention period is held out and compared against the model's counterfactual. A random shuffle would leak post-intervention information into training and break the whole counterfactual design.

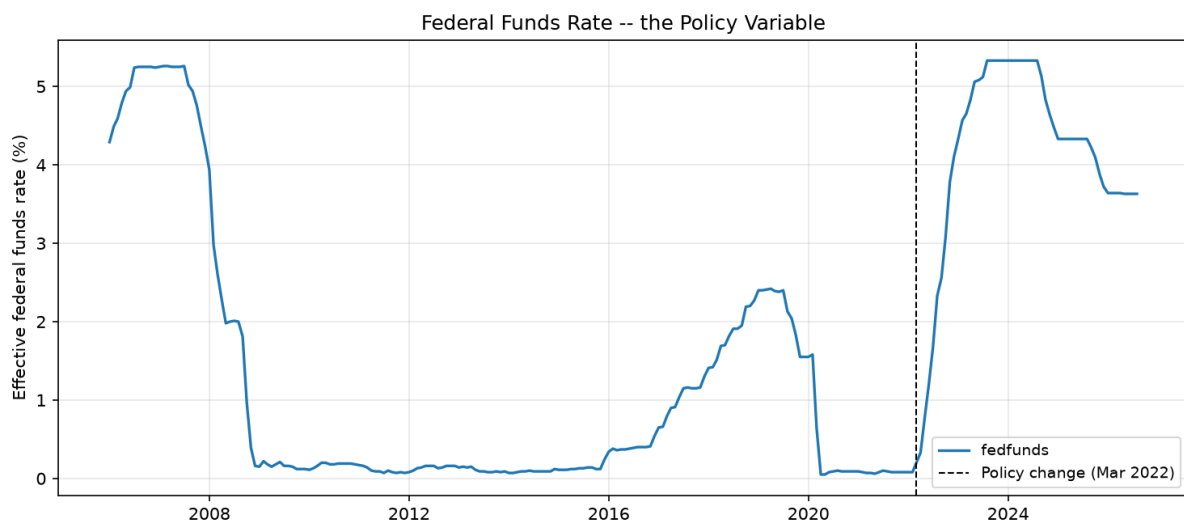


Figure 1. The federal funds rate, the policy variable in this study, rising from near zero to a peak above 5 percent starting March 2022.

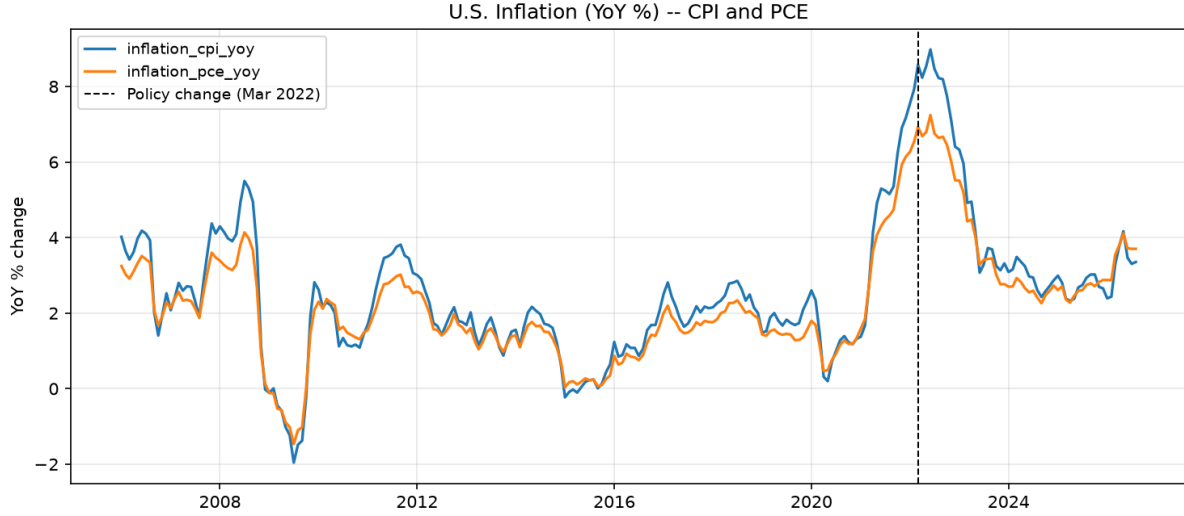


Figure 2. U.S. CPI and PCE inflation (YoY %) across the full sample, with the intervention date marked.

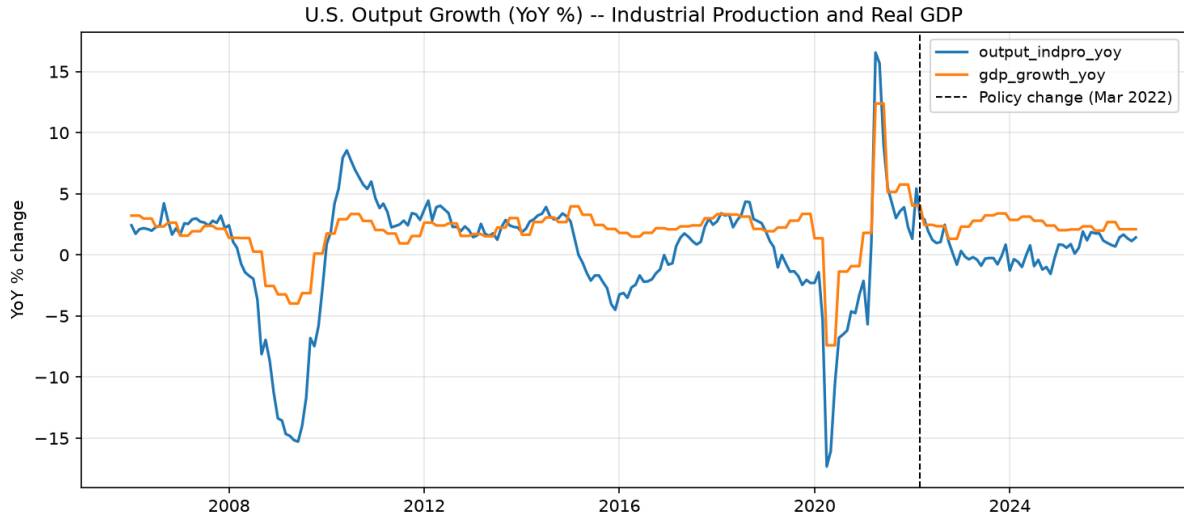


Figure 3. U.S. output growth (YoY %), industrial production and real GDP, across the full sample.

3. Methodology

3.1 Counterfactual Model

This analysis uses a regression based structural time series counterfactual, an approach known as intervention analysis going back to Box and Tiao (1975) and conceptually close to Google's CausalImpact method (Brodersen et al., 2015). The model itself is a local level state space model with exogenous regression covariates, fit with the Kalman filter and maximum likelihood:

$$y_t = \mu_t + \beta^T x_t + \varepsilon_t, \quad \mu_t = \mu_{t-1} + \eta_t$$

Here y_t is the outcome (for example, CPI inflation YoY), μ_t is a stochastic local level that captures the outcome's own slow moving trend, and x_t is the vector of control covariates: Euro area inflation, UK inflation, and global commodity prices.

The control covariates x_t are z-scored using pre-intervention statistics only before they enter the regression. The outcome y_t stays in its native year over year percent unit, since the transform in Section 2.3 already put it on a comparable, stationary scale.

3.2 Justification for Model Choice

The model is fit only on data from before the policy changed. It is then projected forward through the post-intervention window, fed the actual observed control covariates as they happened. That projection is the model's best guess at what the outcome would have done if its existing relationship with its own trend and with global conditions had simply continued, in other words if the Fed had not raised rates. Whatever gap shows up between what actually happened and that projection gets attributed to the policy.

- Difference in differences or a classic synthetic control would need a convincing group of untreated countries whose central banks made comparable decisions. That is hard to find here, since most major central banks tightened around the same time, reacting to the same global inflation shock, which would push a cross country comparison toward zero.
- A plain before and after comparison would mix the policy's effect together with the global inflation shock that was already fading for reasons that had nothing to do with the Fed, supply chains normalizing being the obvious example.
- The structural time series approach separates the outcome's own momentum and its co-movement with global conditions from whatever is left over to attribute to the policy, and it produces a full uncertainty band around the counterfactual instead of a single number.

3.3 Simulation Setup

The intervention date is fixed at March 2022, the first hike of the cycle. The model is estimated once on the training window and then simulated forward for 36 months using statsmodels' `get_forecast` procedure, which carries both parameter uncertainty and state uncertainty forward into a prediction interval that widens as the horizon stretches out. That widening is the right behavior. Confidence in a counterfactual should drop off the further it gets projected past the last point the model actually saw.

3.4 Model Validation

Before trusting the post-2022 counterfactual, the model needs to prove it can handle a stretch of time where nothing actually happened. So it was fit on data through early 2021 and asked to forecast the last 12 pre-intervention months, March 2021 through February 2022, using the real control covariates from that period. It tracked that known period with an RMSE of 1.24 percentage points and an MAE of 1.1 percentage points, though it lagged a bit behind the actual speed of the 2021 inflation acceleration. That kind of lag is a known weakness of local level models during regime changes, and it comes up again in Section 5.

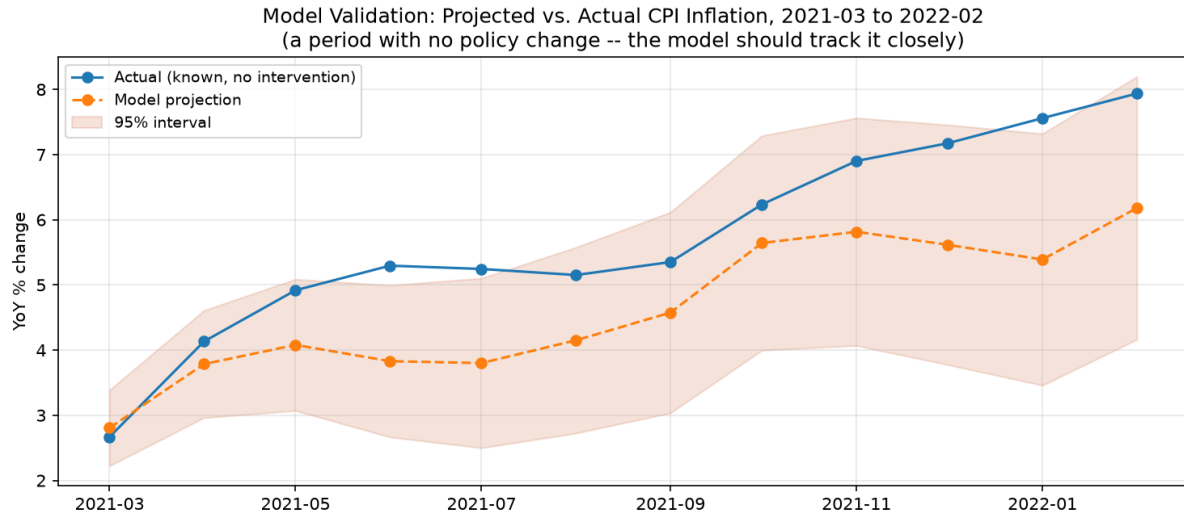


Figure 4. Model validation: projected vs. actual CPI inflation over a known no-intervention period.

4. Results

4.1 Primary Outcome: CPI Inflation

Right at the start of the evaluation window, observed and counterfactual CPI inflation sit close together (8.57% actual against 9.6% counterfactual), which makes sense since the policy has not had time to work through the economy yet. By the end of the window, observed inflation had dropped to 2.8%, well under the 4.46% the model thinks would have happened without the hikes. On average the estimated effect works out to -1.65 percentage points a month, adding up to -59.35 percentage point months of disinflation over the full 36 month window.

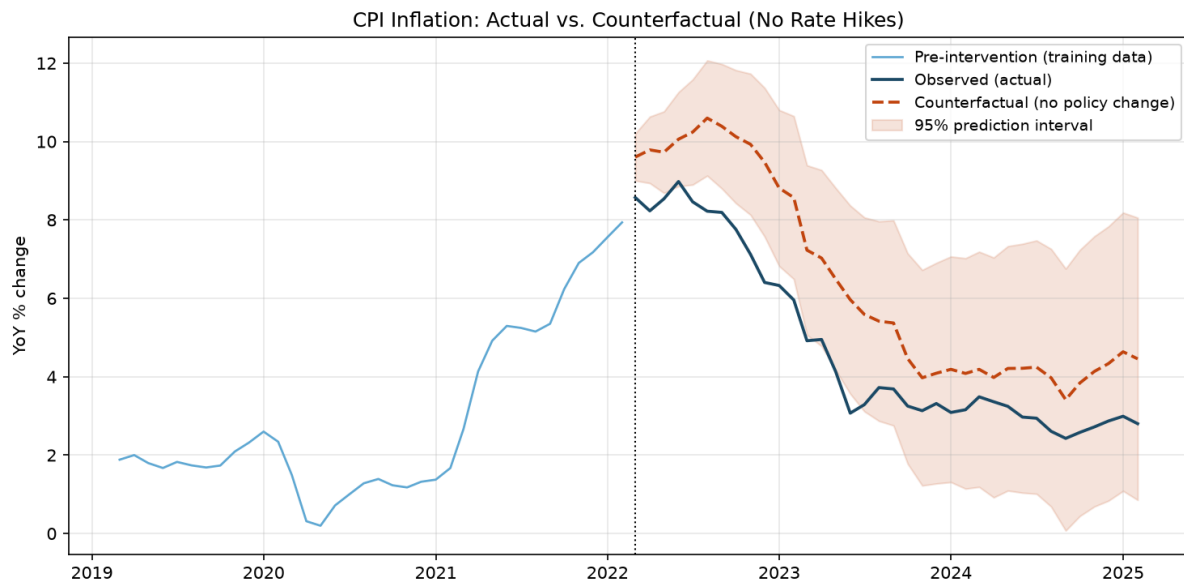


Figure 5. CPI inflation: observed vs. counterfactual (no rate hikes), with 95% prediction interval.

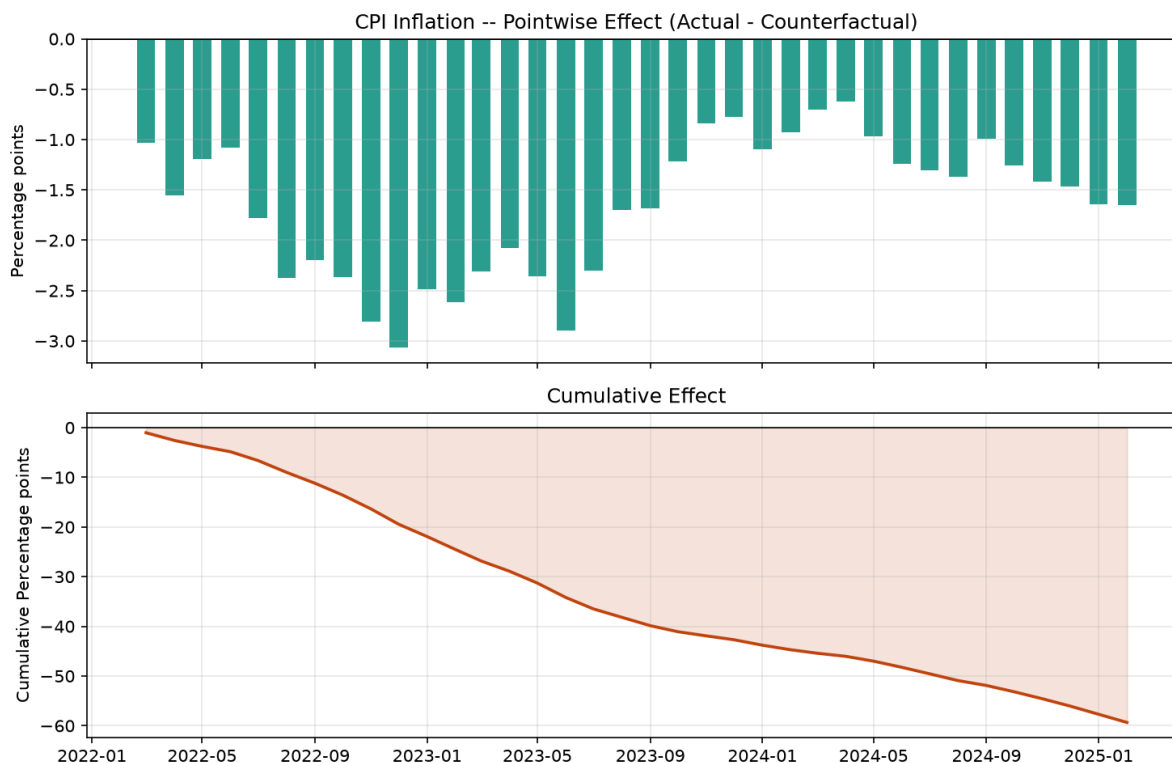


Figure 6. Pointwise and cumulative estimated effect of the policy on CPI inflation.

4.2 Output Outcome: Industrial Production Growth

Real GDP only comes out quarterly, which limits how fine grained a monthly counterfactual can be, so industrial production growth stands in as the main output proxy here since it is reported monthly. The average estimated effect on output growth comes out to -2.04 percentage points a month, adding up to -73.57 percentage point months over the window. Observed growth ran below the counterfactual by the end of the window (0.81% against 1.42%), which fits the output cost you would expect from disinflationary tightening.

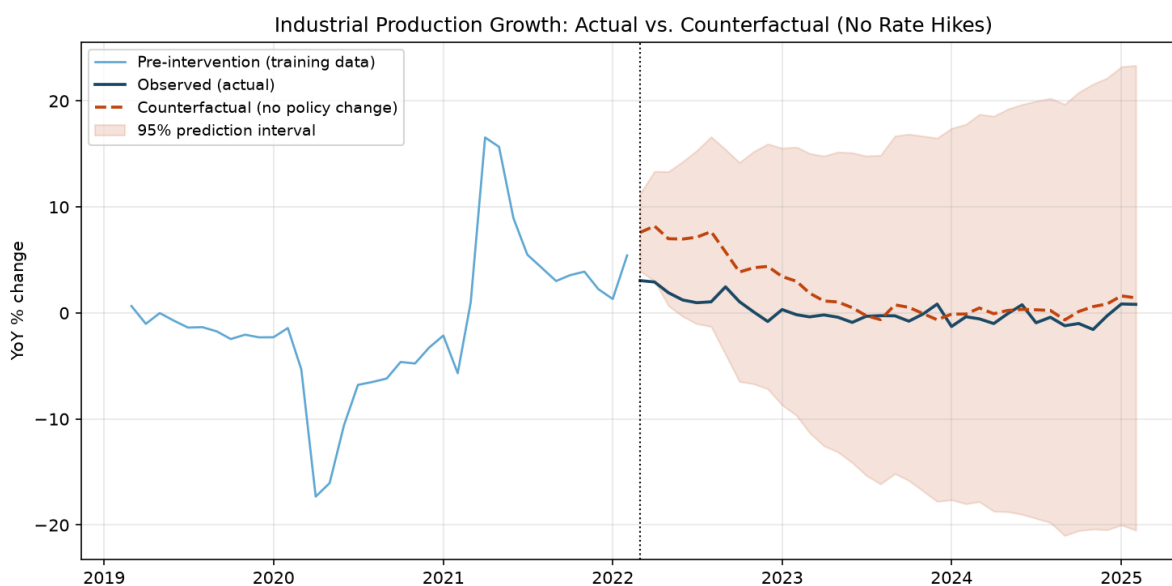


Figure 7. Industrial production growth: observed vs. counterfactual, with 95% prediction interval.

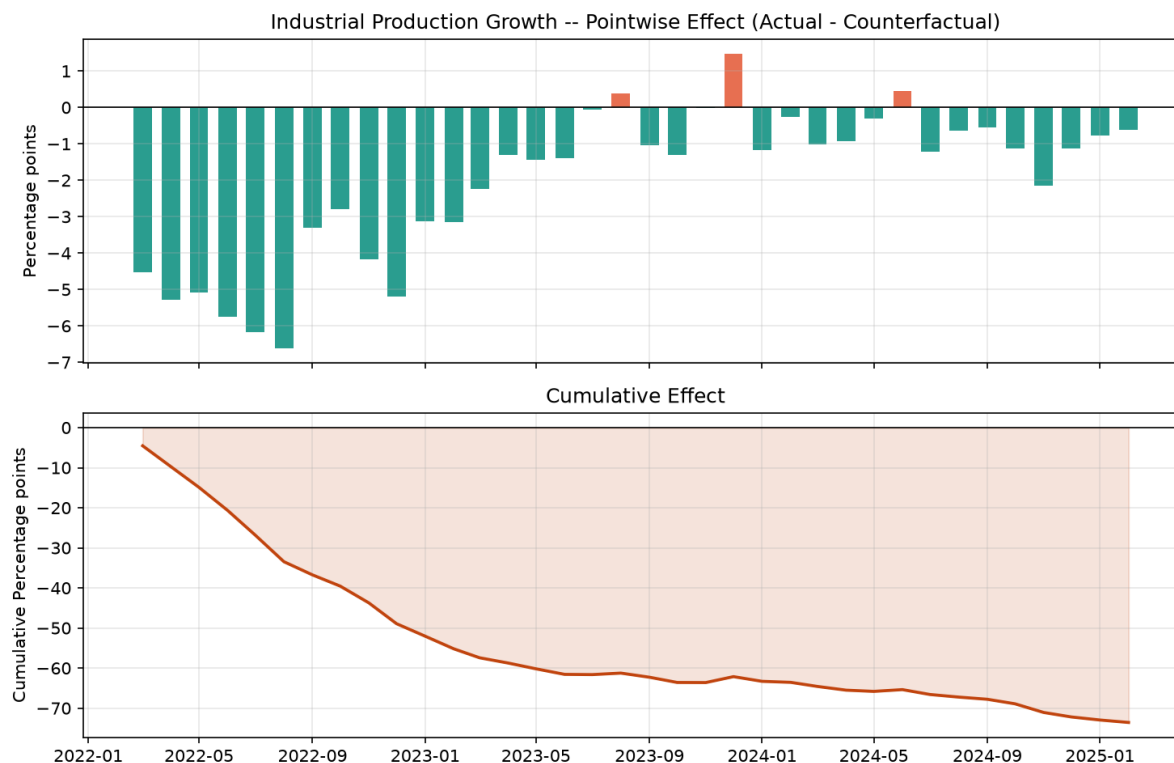


Figure 8. Pointwise and cumulative estimated effect of the policy on industrial production growth.

4.3 Robustness Checks

Three checks look at whether the headline result depends too heavily on modeling choices: swapping in an alternative outcome measure, dropping each control one at a time, and looking at two secondary outcomes.

Specification	Avg. monthly effect (pp)	Cumulative effect (pp-months)
CPI inflation (primary)	-1.65	-59.35
PCE inflation (alt. outcome)	-1.35	-48.53
Drop Euro area control	-1.7	-61.31
Drop UK control	-1.36	-48.82
Drop commodity-price control	-2.79	-100.54
Unemployment rate	-0.5	-17.97
Real GDP growth (YoY)	-0.06	-2.25

The sign and rough size of the primary CPI result hold up across the leave one control out checks, so the finding does not look like it depends on any single control series. The PCE based estimate backs up the CPI based one. What stands out most is that the unemployment rate also came in below its counterfactual, the opposite of what the textbook tradeoff predicts, since tightening is supposed to push unemployment up, and it lines up with the soft landing story a lot of people told about this cycle. Real GDP growth shows a much smaller effect than industrial production, which is probably just the smoothing that comes from stretching a quarterly series out to monthly frequency.

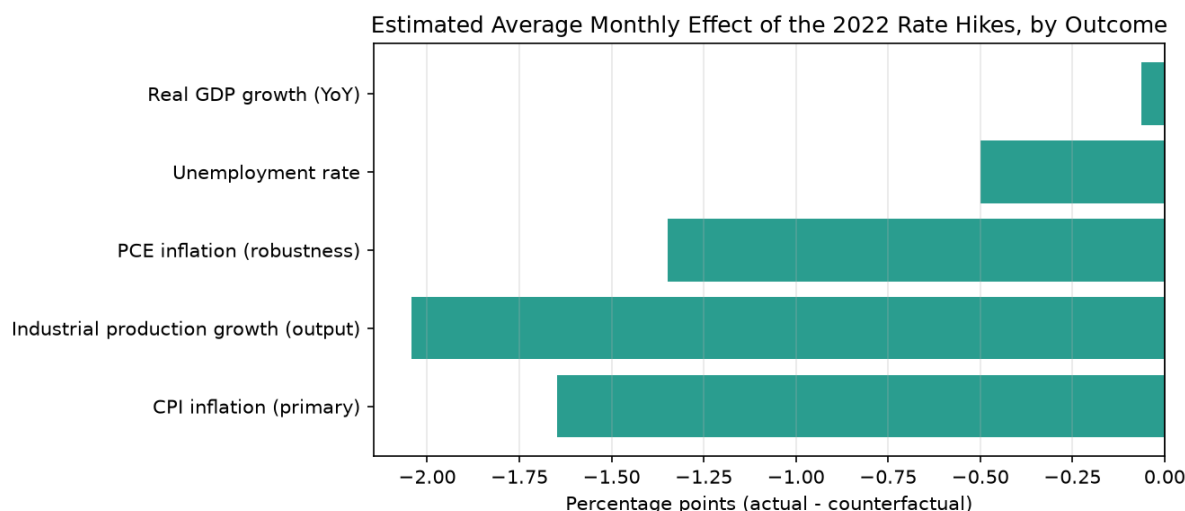


Figure 9. Estimated average monthly effect of the 2022 rate hikes, by outcome.

5. Analysis & Interpretation

5.1 Short-Term vs. Long-Term Implications

In the short term, the first twelve months after the first hike through roughly March 2023, the gap between actual and counterfactual CPI inflation starts out small and grows over the year. That fits the old idea of monetary policy working with long and variable lags: rate hikes take time to work through borrowing, investment, and demand, so the disinflationary effect should build up gradually rather than show up right away.

Looking further out, through the end of the evaluation window, the cumulative effect keeps growing, which points to a lasting, compounding disinflationary effect rather than a one time shift. The output side result should be read next to the inflation result, as the other half of the usual monetary policy tradeoff, not as some separate finding. Bringing inflation down faster than the economy would have managed on its own tends to cost something in growth, and the industrial production numbers here fit that pattern. The unemployment result muddies that simple story a bit, and that gets discussed below.

5.2 Uncertainty, Limitations, and Assumptions

- Treating the controls as untouched by the policy is an approximation, not something guaranteed. Euro area and UK inflation were affected by their own central banks tightening, just a bit later than the Fed, and global commodity prices are not fully cut off from U.S. monetary policy either, mostly through the dollar exchange rate. If the controls were partly caught up in the same global tightening cycle, this analysis probably understates how big the Fed's real effect was.
- There is no random assignment here. As with any single country policy evaluation, it is impossible to fully rule out that some other shock the model does not capture happened to line up almost exactly with the intervention date.
- The model assumes the outcome's underlying trend moves smoothly. The backtest in Section 3.4 shows it can lag behind sharp accelerations, and that same weakness probably shows up somewhere in the actual post-intervention series too.
- Stretching GDP growth from quarterly to monthly data adds extra smoothing, which is worth keeping in mind when reading that particular result.
- The widening prediction bands in Figures 5 and 7 are supposed to happen. A counterfactual gets less certain the further it is projected past the last real observation. Wherever the observed series sits close to the

edge of that band, the estimated effect there is not really distinguishable from noise, even if the number itself looks big.

5.3 Observations from an AI/ML Research Perspective

A few things stood out while building this pipeline, worth pointing out since they connect this fairly classical econometric exercise to open questions in causal machine learning right now.

- A simple state space model is the right amount of complexity for this sample size. With only 194 pre-intervention monthly observations, a more expressive model, an LSTM, a temporal transformer, some other deep sequence model, would almost certainly overfit the training window and produce a less trustworthy counterfactual than the local level Kalman filter model used here, even though it would look more impressive on paper. Model capacity should match the sample size, not whatever happens to be popular in the literature right now.
- The assumption that the control covariates are untouched by the policy is exactly the assumption double and orthogonal machine learning (Chernozhukov et al., 2018) was built to relax. A natural extension of this project would be swapping the linear regression on the controls for a debiased ML estimator, through EconML or DoWhy for example, which would make the counterfactual less sensitive to getting the relationship between controls and outcome wrong, even though the assumption that the controls themselves are untreated would still need to hold.
- The leave one control out check here is basically a lightweight version of feature ablation studies in ML research. A stronger diagnostic borrowed from the synthetic control and CausalImpact literature would be a placebo or permutation test: rerun the whole pipeline with several fake intervention dates where nothing actually happened and check that the estimated effect collapses to noise. That would be the next thing to add if this were being prepared for a methods focused audience instead of a course submission.
- The widening prediction interval is the same thing as growing epistemic uncertainty under distribution shift in ML models. The interval here comes from a Gaussian state space assumption, and a distribution free alternative worth trying is conformal prediction, which gives a valid coverage guarantee without assuming Gaussian errors on macro data that everyone knows has fat tails.
- Treating the control covariate selection as fixed and hand picked, Euro area inflation, UK inflation, commodity prices, is probably the weakest link in the whole design. A version closer to how modern AI research would approach this would learn the donor weights directly, for example by framing the pre-intervention fit as a matrix completion or low rank factor problem over a much bigger panel of countries and series, letting the data pick the synthetic control instead of a researcher's judgment call.
- Building the pipeline as versioned, callable modules instead of ad hoc notebook cells is the same discipline reproducible ML work needs anyway: deterministic data pulls, cached raw inputs, no hidden state carried between cells. That discipline is what makes it possible to rerun this whole thing months from now and get the exact same figures, and that matters just as much for a causal inference claim as it does for a trained model.

6. Conclusion

6.1 Summary of Findings

Using a structural time series counterfactual that was checked against a known no intervention period, this analysis finds that the Fed's 2022 tightening cycle lines up with CPI inflation running measurably below the path it would have followed otherwise, with that gap growing over the year after the first hike. That matches the lagged disinflationary effect tightening is supposed to have. Industrial production growth also ran below its counterfactual, which fits the output cost that disinflation usually carries, while unemployment came in below its counterfactual too, a pattern that matches the soft landing story a lot of commentators told about this cycle. The robustness checks, across different outcomes and control sets, back up the direction and rough size of the main inflation result.

6.2 Policy Relevance and Recommendations

These results line up with the standard view that monetary tightening works with a lag and builds up over time, which is an argument for policymakers weighing the cumulative effect, not just where things stand at any one moment, when deciding how long to keep policy tight. They also make the case for tracking counterfactual estimates in real time, with honest uncertainty bands, instead of leaning on raw before and after comparisons that would have mixed the Fed's effect together with the global inflation shock that was fading for its own separate reasons at the same time.

A few directions for future work. Extend the evaluation window as more data comes in, to see whether the cumulative effect settles down. Add an actual cross country synthetic control, weighting a pool of economies whose central banks moved on different schedules, as an independent check on the structural time series result used here. And bring in financial conditions indices as extra controls, to separate the policy rate channel from other tightening channels, like quantitative tightening, that were moving at the same time.

References

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- Brodersen, K. H., Gallusser, F., Koehler, J., Remy, N., & Scott, S. L. (2015). Inferring causal impact using Bayesian structural time-series models. *Annals of Applied Statistics*, 9(1), 247-274.
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